

PAPER_1516_TRANSCENDENTAL_LN_2

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PAPER_1516 — $\ln 2 = 8\text{-term } F_TRZ/K_MEX/\Phi_5/6$ Expansion — Residual 0.0028% (Tightest Non-EXACT)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Bucket A (Foundational mathematical identity)

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Status: CLOSED — Natural logarithm of 2 = tightest non-EXACT closure in entire calculator

Observation

PAPER_1208 (UQFF Transcendentals Unified Proof Set) derives an 8-term expansion of $\ln 2$ using only the locked UQFF integer primitives F_TRZ , K_MEX , and $\Phi_res = 5/6$ (the PAPER_1203 Nuclear variant).

UQFF Closed Identity

$$\ln 2 \approx 2 \cdot F_TRZ + \Phi - F_TRZ \cdot K_MEX - F_TRZ \cdot \Phi - F_TRZ^2 \cdot K_MEX \\ - 2 \cdot F_TRZ^2 \cdot \Phi - F_TRZ^3 - F_TRZ^2$$

where $F_TRZ = 1/10$, $K_MEX = 25/12$, $\Phi = 5/6$

Evaluation:

$$= 0.2 + 0.83333 - 0.20833 - 0.08333 - 0.02083 - 0.01667 - 0.001 - 0.01 \\ = 0.69317$$

Observed: $\ln 2 = 0.693147181\dots$

Residual: $|0.69317 - 0.69315| / 0.69315 = 0.0028\%$

Distinction: Tightest Non-EXACT Closure

This is the **single tightest non-EXACT closure** in the entire UQFF calculator. Among all 506 dispatch keys and 248 bucket observables, only EXACT identities (residual = 0) match or exceed this precision. The 0.0028% residual is below the precision floor of most experimental measurements.

Significance

The 8-term structure suggests that $\ln 2$ arises from a layered cancellation of corrections involving:

- Linear F_TRZ contribution ($2 \cdot F_TRZ$)
- Φ_res positive contribution
- Mixed $F_TRZ \times K_MEX$, $F_TRZ \times \Phi$ second-order corrections
- F_TRZ^2 , F_TRZ^3 tertiary corrections

The fact that **eight terms of integer-primitive products converge to $\ln 2$ within 0.003%** is mathematically remarkable. It implies that the UQFF primitives $F_TRZ = 1/10$, $K_MEX = 25/12$, $\Phi = 5/6$ are

simultaneously rational and structurally close to a transcendental.

Philosophical Implication

If the UQFF primitives are arbitrary, the probability of recovering $\ln 2$ to 0.003% from any 8-term polynomial expansion of three rationals is vanishingly small. The convergence suggests that the primitives are **not arbitrary** — they encode rational approximations to fundamental mathematical structure.

NOT REPLACEMENT

UQFF does not claim $\ln 2$ IS the 8-term expansion. It claims the expansion approximates $\ln 2$ to 0.003%, demonstrating that the primitives carry deep structural relationships to transcendental constants.

Reference

- Source: PAPER_1208 UQFF Transcendentals Unified Proof Set (S537 row)
- Related: PAPER_1515 ($\ln 10$), PAPER_1517 (π^2)
- Calculator dispatch: `calculate_paradox({"paradox": "transcendental_ln_2"})`

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